Master of Epidemiology · Research Data Management · 2026–27

REDCap training

From your first instrument to a clean, de-identified export

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Based on the official REDCap training videos, projectredcap.org

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GHI | Global Health Institute

Learning objectives

By the end of this session you can:

- **Build** instruments in the Online Designer and with a data dictionary
- **Validate** data at entry: field types, ranges, branching logic, calculations
- **Choose** the right project type, and set up events and repeating forms
- **Run** a multi-site project with Data Access Groups, queries and locking
- **Export** data to R or Stata without identifiers

Six topics, then a hands-on lab

1. Getting started
2. Building instruments
3. Choosing a project type
4. Longitudinal and repeating data
5. Data entry, quality and multi-site work
6. Users, exports and going live

L **Hands-on lab: build TBCARE**

You need

A laptop

Your REDCap login

The TBCARE case study (Part 2)

Fake data only

The official REDCap videos, and where they fit today

Video set	Videos (minutes)	Topic
Introductory overviews	Brief overview (7) · Detailed overview (14)	1
Basic features	Instrument development (6) · Online Designer (5) · Data Dictionary (10) · Basic data entry (19) · Applications overview (16)	2, 5, 6
Project types	Overview (3) · Traditional (3) · Single survey (5) · Longitudinal (3) · Longitudinal + scheduling (3) · Operations (2)	3
Special topics	Field types (4) · Mobile App (2) · Calendar (7) · Data Access Groups (7) · Data queries (5) · Locking records (2) · Defining events (5) · Designating instruments (3) · Scheduling (7) · Repeating instruments and events (33)	2, 4, 5



Before the lab, watch the **two overviews** (21 min). Videos may show an older REDCap version.

Roadmap

1. **Getting started**
2. Building instruments
3. Choosing a project type
4. Longitudinal and repeating data
5. Data entry, quality and multi-site work
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TOPIC 1 OF 6

Getting started

In this topic

- What REDCap is, and how it supports your DMP
- The project lifecycle: development to production
- Creating a project
- The Project Setup checklist

What is REDCap?

8,400+ institutions

166 countries

2.8 M projects

4.6 M users

A secure web application for **surveys and databases**. Free for non-profits; **hosted by your own institution**.

Secure
audit trail, user rights

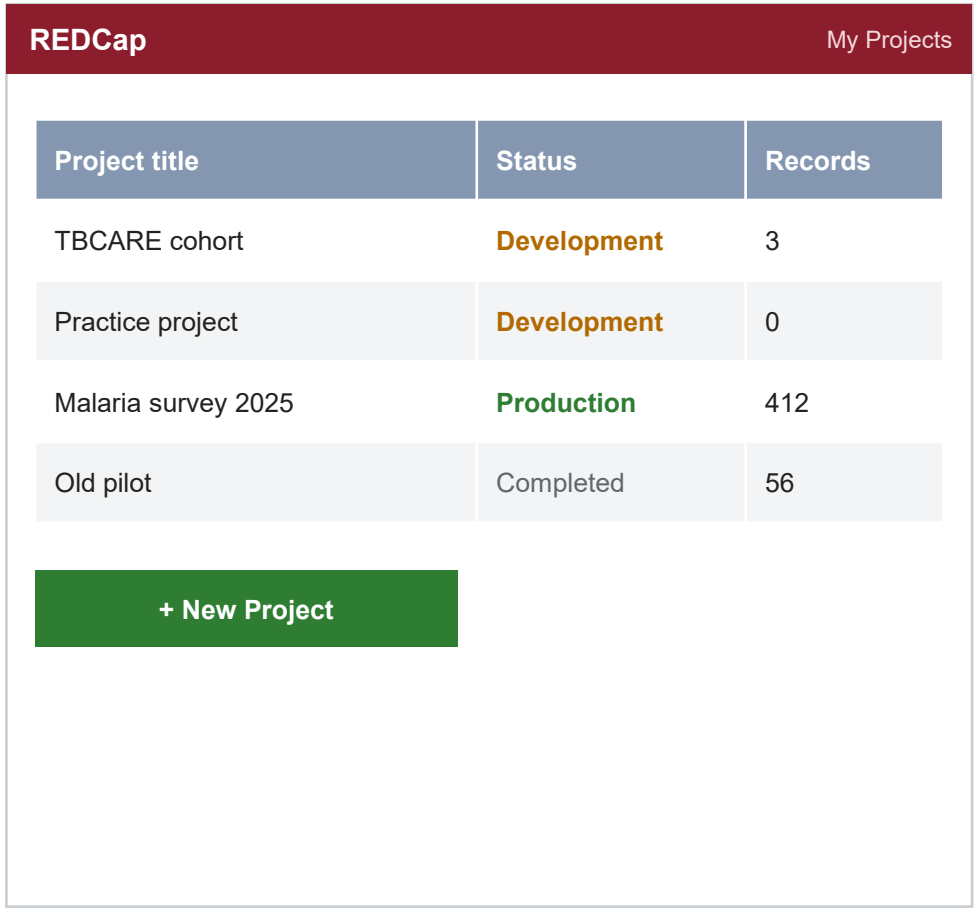
Forms and surveys
no programming

Longitudinal
events, repeating forms

Multi-site
Data Access Groups

Offline
Mobile App, MyCap

Exports
R, Stata, SPSS, SAS, API



REDCap: Research Electronic Data Capture · SPSS: Statistical Package for the Social Sciences · SAS: Statistical Analysis System · API: application programming interface

WATCH
Brief overview of REDCap (7 min)

REDCap puts your DMP into practice

DMP section	What you promised	How REDCap delivers it
2 Documentation & quality	A codebook; checks at entry	Data dictionary; validation; Data Quality rules; queries
3 Storage & backup	Secure institutional storage	Institutional server; backups by the host; user rights
4 Legal & ethical	Personal data protected	Identifier flags; Data Access Groups; logging (audit trail)
5 Sharing	De-identified sharing	De-identified exports: remove identifiers, shift dates, hash IDs
6 Responsibilities	Named roles	User roles and rights; e-signatures; locking

✓ In your DMP, name the **REDCap feature**, not just “a secure database”.

The project lifecycle: test first, then go live



Build and test with fake data

Real data; changes via Draft Mode

Collection done; clean and export

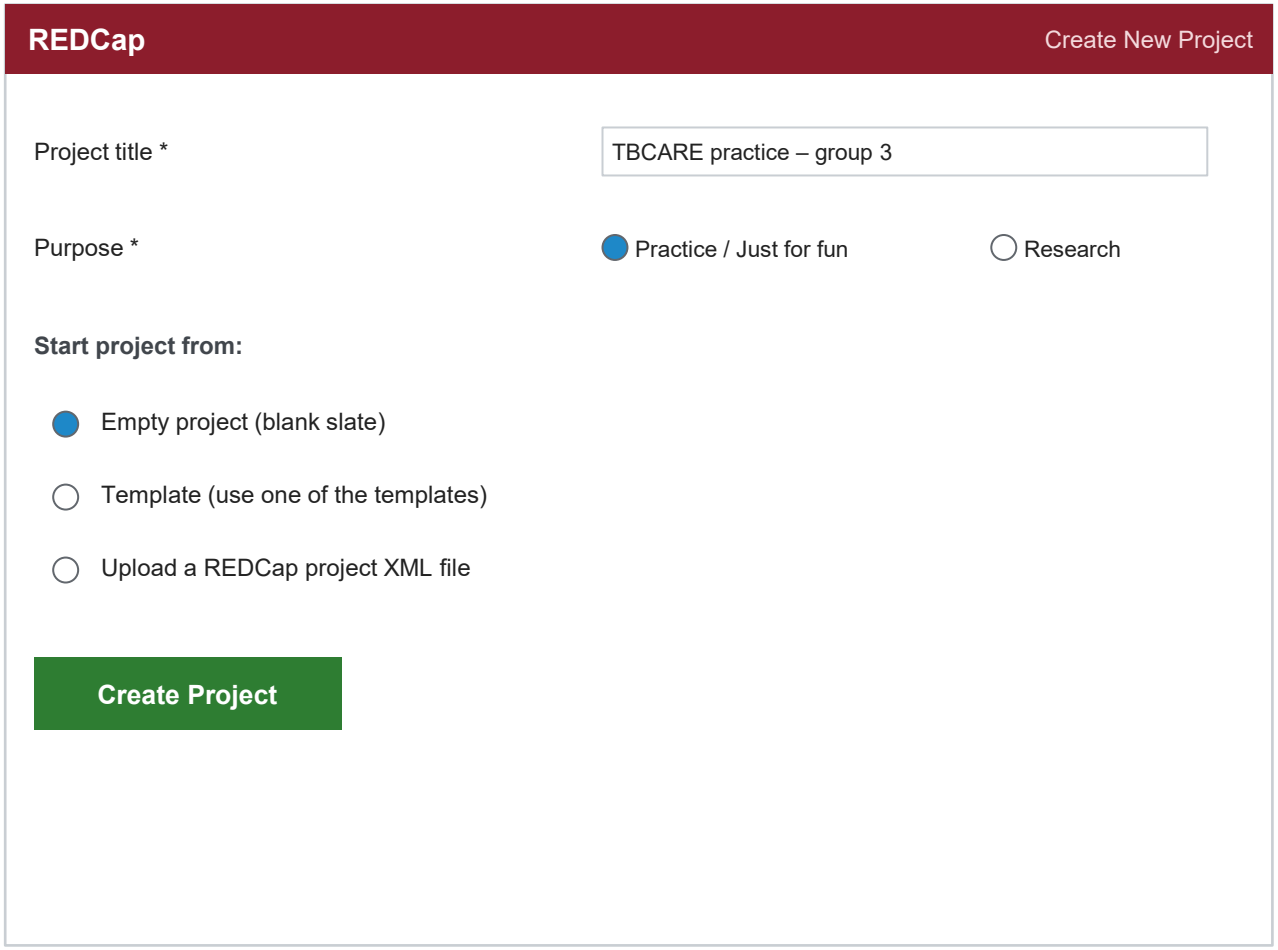
Read-only archive

⚠ **Never enter real data in Development.** Changes in Production can damage collected data.

✓ Test every field and branch with a colleague **before** you move to Production.

WATCH
Detailed overview of REDCap (14 min)

Creating a project



Empty build from scratch

Template start from an example

XML upload copy a whole project design

For this course, purpose = **Practice / Just for fun**.

WATCH
Detailed overview of REDCap (14 min)

The Project Setup page is your checklist

- ✓ **Main project settings** · surveys? longitudinal?
- ✓ **Design your instruments** · Online Designer or Data Dictionary
- ○ **Define events (longitudinal)** · and designate instruments
- ○ **Optional modules** · repeating, auto-numbering, scheduling, queries
- ○ **User rights and DAGs** · who sees what
- ○ **Test** · with fake records
- ○ **Move to Production** · after testing

Two switches first

Use **surveys**?
Use **longitudinal** events?
They change the whole project

WATCH
Detailed overview of REDCap (14 min)

Roadmap

1. Getting started
2. **Building instruments**
3. Choosing a project type
4. Longitudinal and repeating data
5. Data entry, quality and multi-site work
6. Users, exports and going live

TOPIC 2 OF 6

Building instruments

In this topic

- Records, instruments and fields
- The Online Designer
- The data dictionary (CSV)
- Field types and validation
- Branching logic, calculations, action tags
- Design rules

CSV: comma-separated values

Records, instruments and fields

PROJECT: TBCARE

Enrolment	Clinical	Laboratory
INSTRUMENT	INSTRUMENT	INSTRUMENT
record_id	weight_kg	lab_eval
dob	height_cm	test_type
sex	bmi	smear

each box = one **field** (variable)

Record — one participant, identified by record ID

Instrument — a form or a survey

Field — one variable or question

Online Designer

point and click

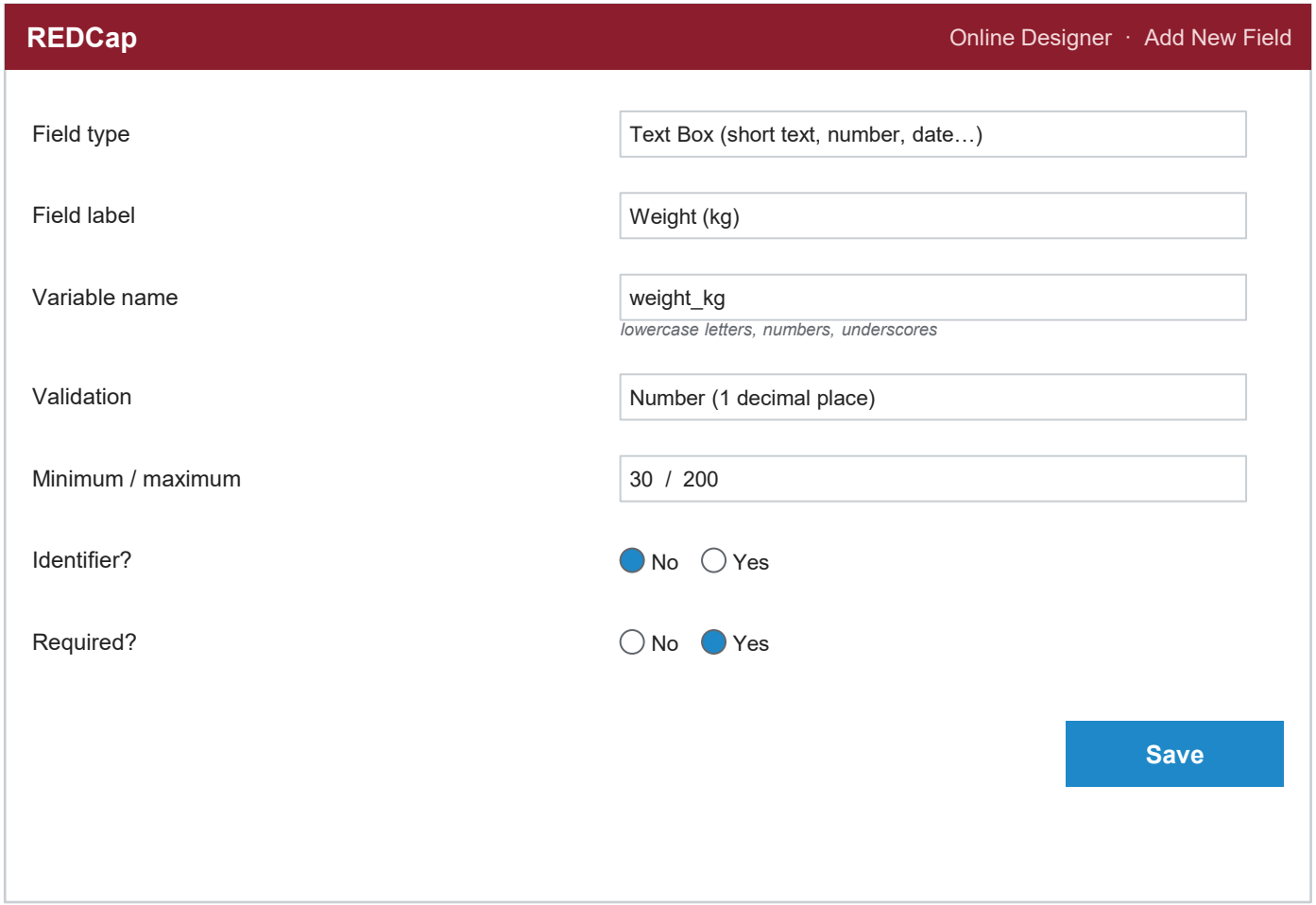
Data Dictionary

CSV, one row per field

WATCH
Introduction to instrument development (6 min)

ID: identifier · CSV: comma-separated values

The Online Designer: add a field



- Choose the **field type**
- Short, clear **variable name**
- **Validation** with min and max
- Tick **Identifier** for personal data
- Mark key fields **required**

⚠ Variable names are **permanent** once data exist.

▷ **WATCH**
The Online Designer (5 min)

The data dictionary: your project as a spreadsheet

Variable / Field Name	Form Name	Field Type	Field Label	Choices, Calculations…	Validation	Min	Max	Identifier?	Branching Logic
record_id	enrolment	text	Study ID						
dob	enrolment	text	Date of birth		date_dmy			y	
sex	enrolment	radio	Sex at birth	1, Male | 2, Female					
weight_kg	clinical	text	Weight (kg)		number	30	200		
height_cm	clinical	text	Height (cm)		number	100	220		
bmi	clinical	calc	BMI	round([weight_kg]/([height_cm]/100)^2,1)					
lab_eval	laboratory	yesno	Lab evaluation done?						
test_type	laboratory	checkbox	Test type	1, Smear | 2, GeneXpert					[lab_eval] = '1'



Design the dictionary **first**: it is also the **codebook** for your DMP.

ID: identifier · BMI: body mass index · DMP: Data Management Plan

WATCH
The Data Dictionary (10 min)

Field types: the right type prevents errors

Text box
with validation

Notes box
long free text

Radio
one answer

Drop-down
one from a long list

Checkboxes
several answers

Yes / No
coded 1 / 0

Calculated
computed for you

Slider
visual scale 0–100

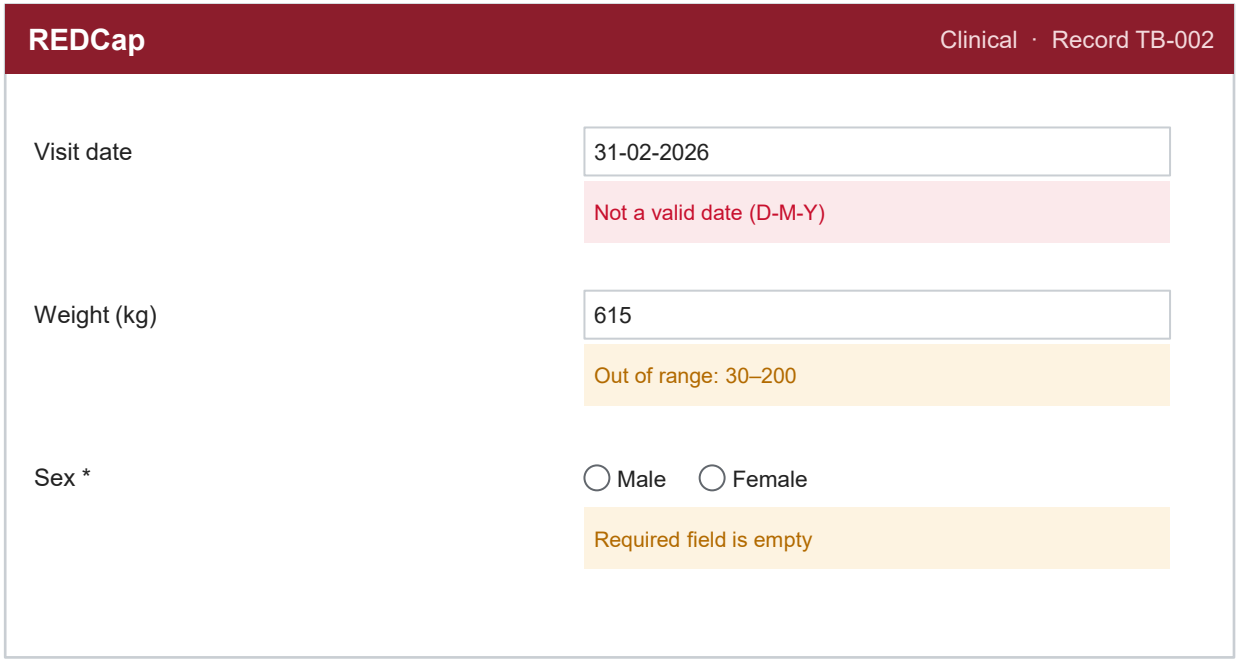
File / signature
uploads

Descriptive
instructions only

Fixed list of answers? **Never use free text.**

WATCH
Project field types (4 min)

Validation: stop errors at the moment of entry



Validation	Example
date_dmy	14-03-1988
integer	3
number_1dp	61.5
email	x@uantwerpen.be
phone	+32 3 …

Set **min and max** for every number and date. Tick **Identifier** for names, dates of birth, phone numbers.

WATCH
Project field types (4 min)

Branching logic: show a question only when it applies

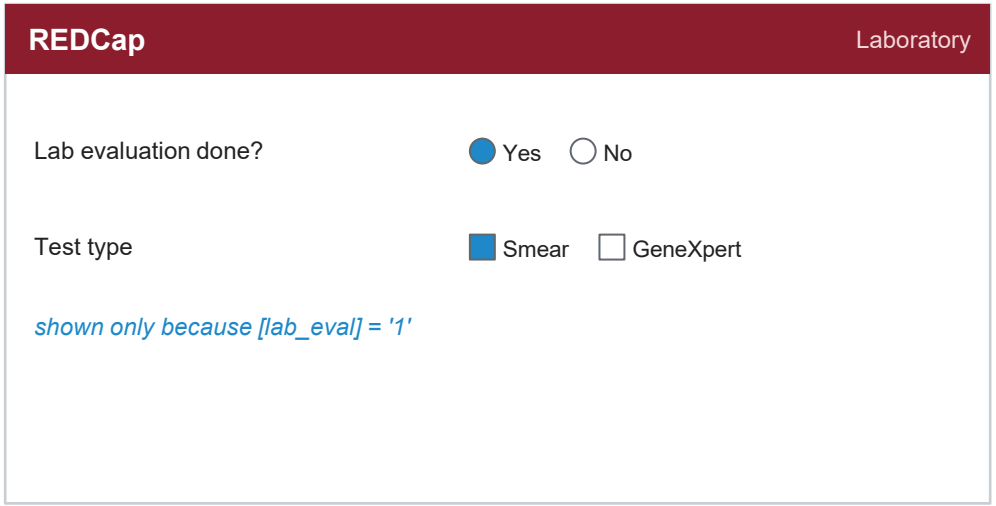
BRANCHING LOGIC EXAMPLES

```
# only if a lab test was done
[lab_eval] = '1'

# women aged 15–49
[sex] = '2' and [age] >= 15 and [age] <= 49

# checkbox option 3 ticked
[symptoms(3)] = '1'

# value from another event
[baseline_arm_1][hiv_status] = '1'
```



{ } **[variable] = 'code'**: brackets for names, quotes for values.

WATCH
Introduction to instrument development (6 min)

Calculated fields, action tags and field embedding

CALCULATED FIELDS

```
# BMI, 1 decimal
round([weight_kg]/([height_cm]/100)^2, 1)

# age in years at enrolment
rounddown(datediff([dob], [enrol_date], 'y'), 0)
```

Action tags

`@TODAY`	today's date	`@HIDDEN`	hide the field
`@READONLY`	no editing	`@DEFAULT='…'`	default value
`@CALCTEXT`	calculated text	`@NONEOFTHEABOVE`	exclusive checkbox

Field embedding: {weight_kg} inside a label or table

Weight (kg)	Height (cm)	BMI
{weight_kg}	{height_cm}	{bmi}

Looks like a paper CRF table; each { } is a real field.

BMI: body mass index · CRF: case report form

Design rules that save you weeks later

✓ Do

- Short, lowercase variable names
- Fixed lists for categorical answers
- Units in the label: “Weight (kg)”
- Min and max for every number and date
- Missing data codes (NA, UNK)
- Test every branch with fake records

✗ Don't

- Free text for categories
- Change a code's meaning after data exist
- Rename variables in Production
- Put names in the record ID
- Collect what the protocol does not need



Most cleaning problems start in **instrument design**. Fix them before you collect.

ID: identifier

Roadmap

1. Getting started
2. Building instruments
3. **Choosing a project type**
4. Longitudinal and repeating data
5. Data entry, quality and multi-site work
6. Users, exports and going live

TOPIC 3 OF 6

Choosing a project type

In this topic

- Five project types
- Which one fits your study?
- Traditional and single-survey projects
- Longitudinal, scheduling and operations projects

Five project types

Traditional

Forms entered by staff, one time point

video 3 min

Single survey

One survey, completed by participants

video 5 min

Longitudinal

Repeated visits (events) per record

video 3 min

Longitudinal + scheduling

Visits with planned dates and windows

video 3 min

Operations

Non-research tracking and administration

video 2 min

They are combinations of two switches on Project Setup:

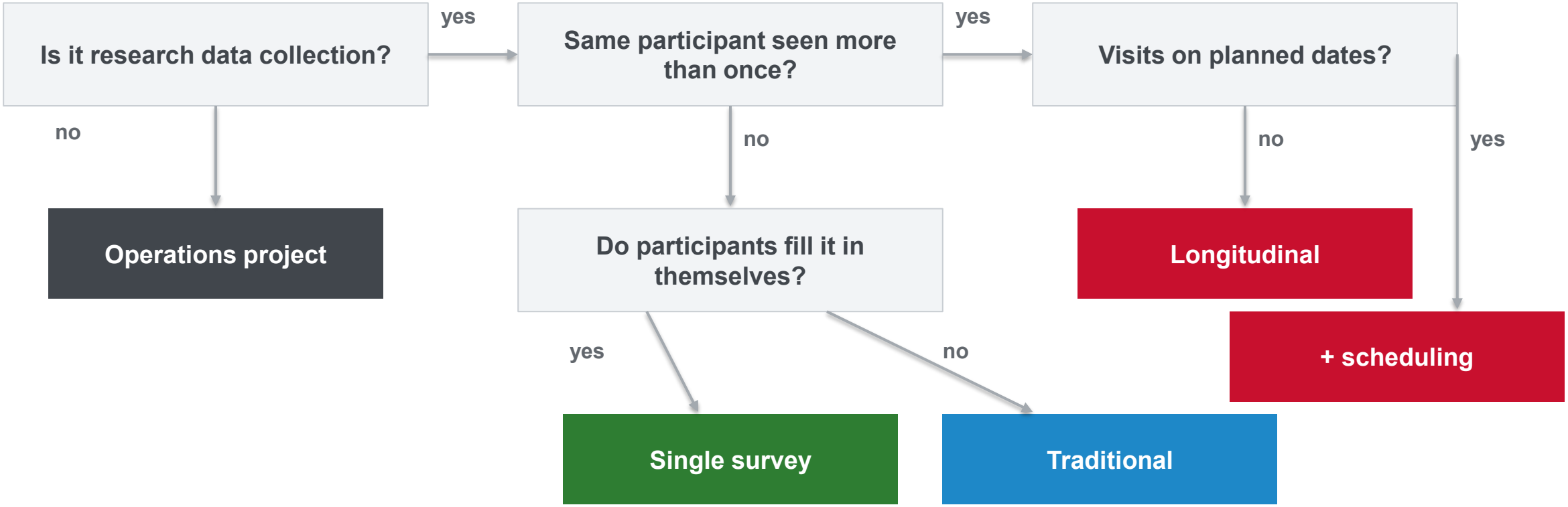
Surveys on / off

Longitudinal on / off

+ Scheduling module

WATCH
Overview of project types (3 min)

Which project type fits your study?



TBCARE: **longitudinal** (4 visits), with surveys on for the quality-of-life form.

Traditional and single-survey projects

Traditional

Staff enter data on forms

One time point

e.g., cross-sectional survey, chart review

Record Status Dashboard shows progress

Single survey

Participants answer online

Public link or QR code

e.g., anonymous staff survey

Public link = no identity check

Identified participants? Use the **participant list**, not the public link.

QR: quick response (code)

WATCH
Traditional project (3 min) · Single survey project (5 min)

Longitudinal, scheduling and operations projects

Longitudinal

- Events = visits
- Same form at several events
- Cohorts, trials, follow-up

+ Scheduling

- Planned dates and windows
- A schedule per participant
- Shown in the Calendar

Operations

- Not research data
- Tracking, registers, requests
- e.g., sample shipments

Switch on longitudinal **before** collecting data: changing later is complex.

WATCH
Longitudinal project (3 min) · Longitudinal + scheduling (3 min) · Operations project (2 min)

Roadmap

1. Getting started
2. Building instruments
3. Choosing a project type
4. **Longitudinal and repeating data**
5. Data entry, quality and multi-site work
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TOPIC 4 OF 6

Longitudinal and repeating data

In this topic

- Arms and events
- Designating instruments for events
- Repeating instruments and repeating events
- Scheduling and the Calendar

Arms and events

REDCap — Define My Events · Arm 1

Event	Days offset	Window (–/+)	Unique event name
Baseline	0	0 / 0	baseline_arm_1
Month 2	60	7 / 7	month_2_arm_1
Month 6	180	14 / 14	month_6_arm_1
Month 12	365	14 / 14	month_12_arm_1

Event

a time point, e.g. Month 2

Arm

a group with its own schedule
most studies: one arm

{ } Unique event names (**baseline_arm_1**) appear in logic and exports.

WATCH
Defining longitudinal events (5 min)

Designate instruments: what is collected when

	Baseline	Month 2	Month 6	Month 12
Enrolment	✓	●	●	●
Clinical	✓	✓	✓	✓
Laboratory	✓	✓	✓	✓
Quality of life (survey)	✓	●	✓	✓
Treatment outcome	●	●	✓	✓

In REDCap

Project Setup
Define My Events
Designate Instruments
Tick the grid

✓ This grid becomes each participant's **Record Status Dashboard**.

WATCH
Designating instruments for events (3 min)

Repeating instruments and repeating events

Repeating instrument

One form, as often as needed

e.g., each medication, each adverse event

Within an event

Repeating event

A whole visit, repeated

e.g., unscheduled visits

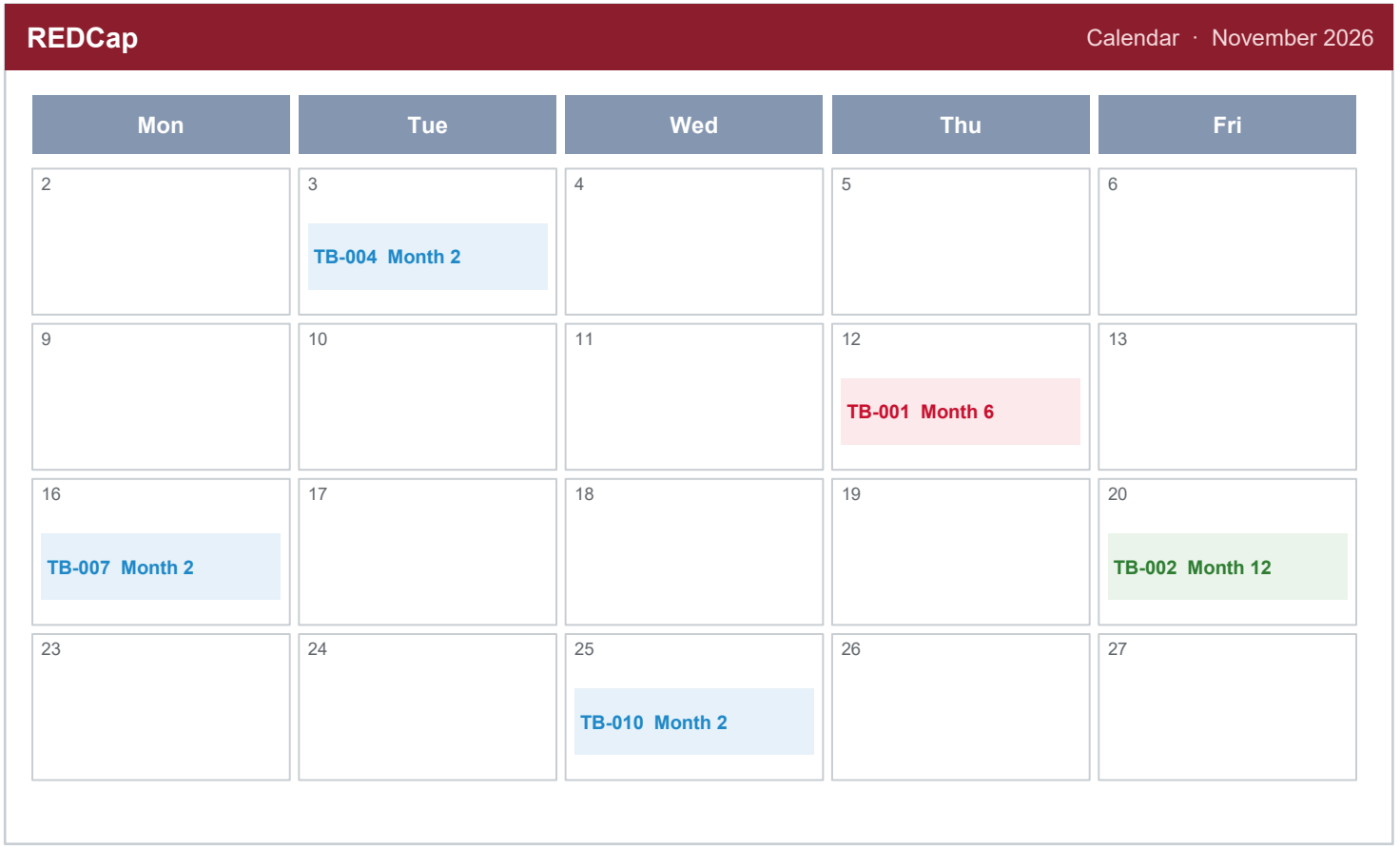
All forms of that event

In the export, each repeat is a row: **redcap_repeat_instrument** and **redcap_repeat_instance**.

✓ Unknown number of occurrences? **Repeat**; do not create med_1, med_2, med_3 fields.

WATCH
Repeating instruments and events (33 min)

Scheduling and the Calendar



Schedule from each event's offset and window

Calendar who is due this week

Status attended, cancelled, no-show

Windows come from **Define My Events**: set them carefully.

WATCH
Scheduling module (7 min) · The Calendar (7 min)

REDCap: Research Electronic Data Capture · TB: tuberculosis

Roadmap

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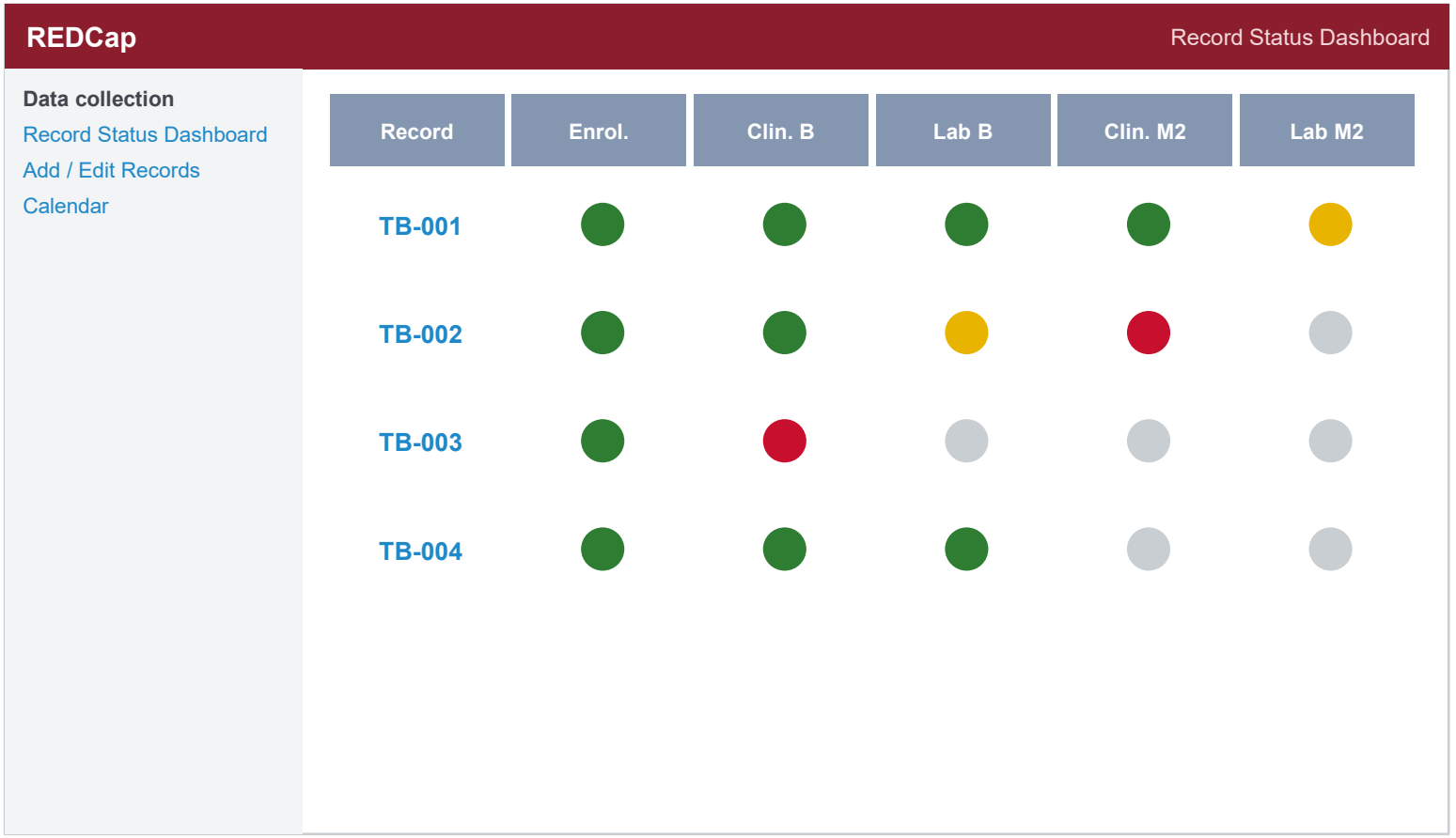
TOPIC 5 OF 6

Data entry, quality and multi-site work

In this topic

- Basic data entry and the Record Status Dashboard
- Surveys and participants
- Data Access Groups
- Queries (Data Resolution Workflow)
- Data Quality rules and locking
- The Mobile App

Basic data entry: the Record Status Dashboard



- Complete
- Unverified
- Incomplete
- No data

Agree as a team what **green** means (e.g., checked against the source).

WATCH
Overview of basic data entry (19 min)

REDCap: Research Electronic Data Capture · TB: tuberculosis

Surveys: let participants enter their own data

Public link and QR code

one link for everyone

Participant list

personal links, reminders

e-Consent

signed PDF with version and date

Alerts and notifications

e.g., e-mail when an event is reported

Multi-Language Management

one form, many languages

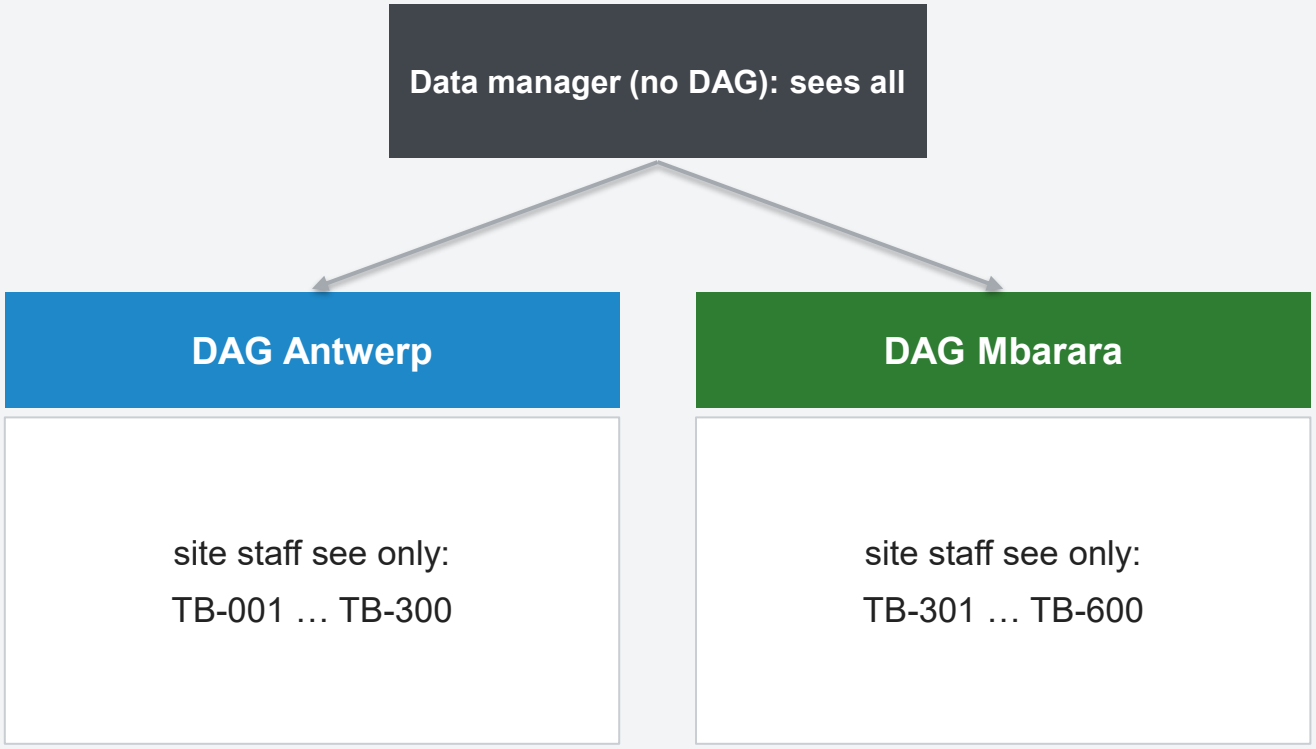
Survey queue

several surveys in a row

Electronic consent? Check that your **ethics committee** accepts it.

QR: quick response (code)

Data Access Groups for multi-site projects



One DAG per site

Sites cannot see each other

New records join the user's DAG

TBCARE: **two DAGs**, Antwerp and Mbarara.

DAG: Data Access Group · TB: tuberculosis

WATCH
Data Access Groups for multi-site projects (7 min)

Queries: the Data Resolution Workflow

1 → 2 → 3

Open

data manager asks: “Weight 300 kg?”

Respond

site corrects to 60.0 or confirms

Close

data manager closes the query

Every step is **time-stamped**: the audit trail for monitors and auditors.

Use the **Resolve Issues** page to see all open queries before lock.

WATCH
Data Resolution Workflow – data queries module (5 min)

Data Quality rules: find errors that slipped through

Built-in checks (examples)	
Missing values	fields left blank
Out of range	values outside min and max
Outliers	extreme numeric values
Calculations	stored value ≠ current formula

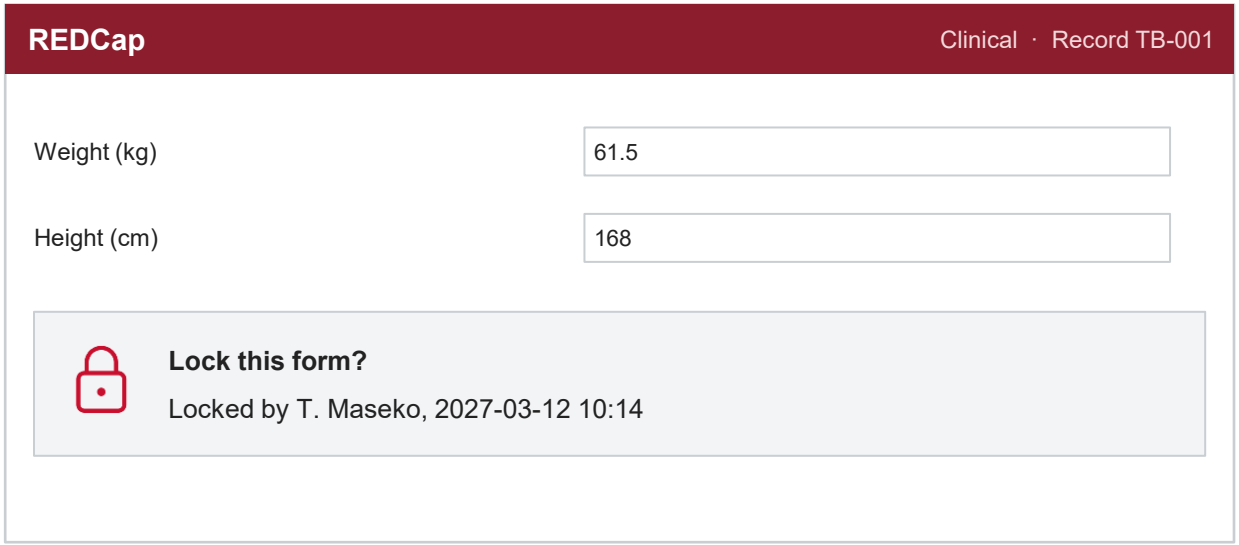
CUSTOM RULES

```
# pregnant men
[sex] = '1' and [pregnant] = '1'

# visit before enrolment
datediff([enrol_date],[visit_date],'d',true) < 0
```

Run the rules **weekly** and turn each problem into a **query**.

Locking records and e-signatures



Lock no more changes

E-signature confirms correctness

Logged unlocks are recorded

Lock after **source verification**; database lock comes after **all queries are closed**.

WATCH
Locking records (2 min)

REDCap: Research Electronic Data Capture · TB: tuberculosis

Collecting offline: the REDCap Mobile App and MyCap

REDCap Mobile App

For **staff**

Collect offline on a tablet

Synchronise when back online

Rural sites, home visits

MyCap

For **participants**

Patient-reported outcomes

Reminders on their phone

Active tasks

Data stay on the device until synced: **lock the device and sync often**.

WATCH
REDCap Mobile App (2 min)

Roadmap

1. Getting started
2. Building instruments
3. Choosing a project type
4. Longitudinal and repeating data
5. Data entry, quality and multi-site work
6. **Users, exports and going live**

TOPIC 6 OF 6

Users, exports and going live

In this topic

- The Applications menu
- User rights and roles
- Logging: the audit trail
- Reports and de-identified exports
- The API
- Moving to Production
- AI tools in REDCap

API: application programming interface · AI: artificial intelligence · REDCap: Research Electronic Data Capture

The Applications menu

- Data Exports, Reports and Stats
- Data Import Tool
- Logging
- User Rights
- Data Access Groups
- Data Quality
- Calendar
- File Repository
- API
- Data Comparison Tool
- Field Comment Log
- Record locking

What each user sees depends on their **user rights**.

API: application programming interface

WATCH
Applications overview (16 min)

User rights: give each role only what it needs

Role	Enter / edit	Export	Queries	Design	User rights
Site data entry	✓	–	respond	–	–
Monitor	view	de-identified	open	–	–
Data manager	✓	full	✓	✓	–
Statistician	–	de-identified	–	–	–
Principal investigator	view	full	✓	–	✓



Least privilege; set **expiry dates**; remove leavers the same day.



WATCH
Applications overview (16 min)

Logging: every change is recorded

REDCap — Logging

Time	User	Action	Changes
2026-11-03 09:12	site_mbr1	Update record TB-304	weight_kg = '615'
2026-11-03 14:40	dm_tbcare	Open query TB-304	“Weight 615 kg?”
2026-11-04 08:05	site_mbr1	Update record TB-304	weight_kg = '61.5'
2026-11-05 10:30	stat_ua	Data export	de-identified, CSV

Who

When

What changed

Exports too

Logging is your **audit trail**: name it in DMP section 2.

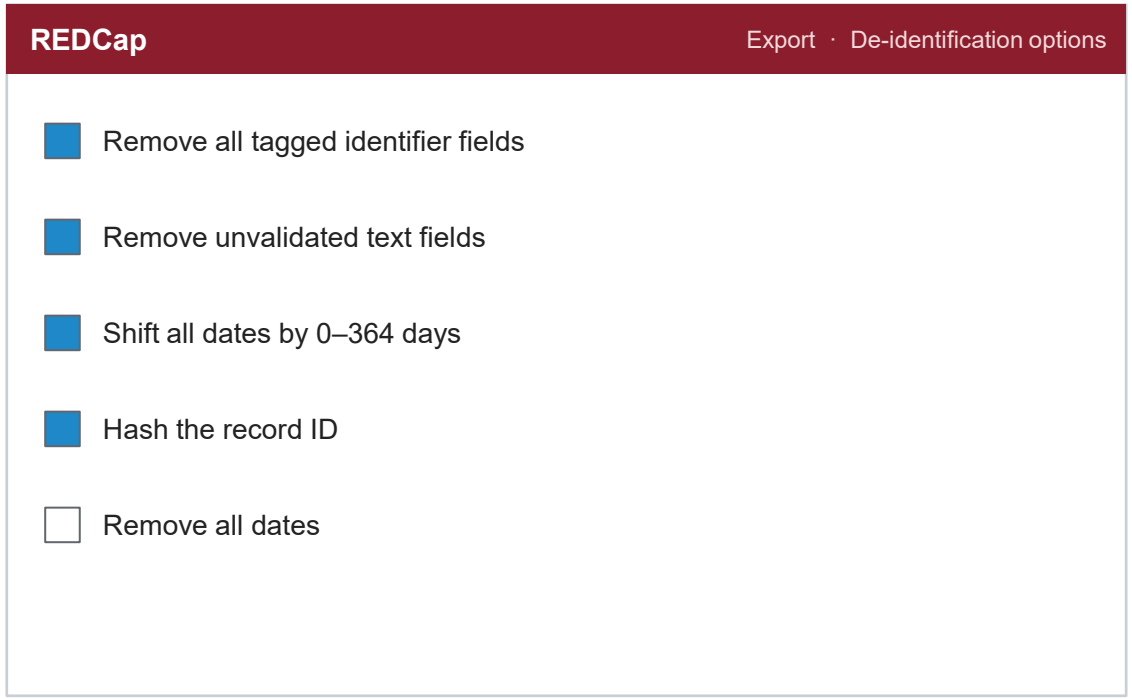
REDCap: Research Electronic Data Capture · TB: tuberculosis · CSV: comma-separated values · DMP: Data Management Plan

Reports and de-identified exports

Export formats

CSV (raw)	CSV (labels)	R	Stata
SPSS	SAS	CDISC ODM (XML)	

Reports: choose fields and filters,
e.g., “Month 2 lab results, Mbarara only”.



⚠ De-identification works only if you ticked **Identifier** when building.

CSV: comma-separated values · SPSS: Statistical Package for the Social Sciences · SAS: Statistical Analysis System · CDISC: Clinical Data Interchange Standards Consortium · ODM: Operational Data Model · XML: Extensible Markup Language · REDCap: Research Electronic Data Capture · ID: identifier

WATCH
Applications overview (16 min)

The API: reproducible data pipelines

R · REDCapR

```r
library(REDCapR)

tb <- redcap_read_oneshot(
  redcap_uri = "https://redcap.uantwerpen.be/api/",
  token      = Sys.getenv("REDCAP_TOKEN")
)$data
```

API token = password

Keep it in **.Renviron**

Never in scripts, GitHub, e-mail

Never in an AI chat

One token per user and project

{ } Python users: the **PyCap** package does the same.

API: application programming interface · AI: artificial intelligence

Before you move to Production: checklist

- ✓ Every instrument tested with fake records
- ✓ Every branch and calculation checked
- ✓ Identifier flags set on personal fields
- ✓ Validation, min and max on numbers and dates
- ✓ Events and instrument designation final
- ✓ User roles and DAGs set up
- ✓ Test records deleted
- ✓ DMP and data dictionary up to date

Ask a colleague who did **not** build it to test it.

DMP: Data Management Plan

AI tools inside REDCap (version 15)

Writing Helper	Summarize	Auto-translate
tone, length, reading level	free-text answers in a report	Multi-Language Management

- Off by default: **the institution decides**
- A human reviews every output
- Never changes research data directly

Not available on your server? The rest of REDCap works the same.

AI: artificial intelligence · REDCap: Research Electronic Data Capture

EXTENSION · TBCARE

From paper CRF to analysis in R

- The database development roadmap
- TBCARE design: two arms, repeating events
- The data dictionary and build order
- Uploading simulated data
- Record linking with the national TB register
- R: REDCapR, REDCapDM, REDCapSync and RosyREDCap

On the website

Slides and CRFs
Data dictionary
Simulated data
R scripts
R you can run in the browser

bosangir.github.io/REDCapMepi/coursework

CRF: case report form · TB: tuberculosis · CRFs: case report forms

The roadmap of a REDCap database



1 **Protocol and DMP**

what and why

2 **CRFs**

paper and annotated

3 **Data dictionary**

CRFs as a spreadsheet

4 **Build**

Development status



5 **Test**

fake data, second person

6 **Production**

after sign-off

7 **Collect and clean**

queries, rules, monitoring

8 **Lock and analyse**

de-identified export, R

⚠ A change after Production goes back to **CRF, dictionary and test** (Draft Mode).

REDCap: Research Electronic Data Capture · DMP: Data Management Plan · CRFs: case report forms · CRF: case report form

TBCARE design: two arms, repeating events

Arm 1 · drug-susceptible TB · 6 months

Form	B	M2	M6	M12	Uns	Log
Screening	✓					
e-Consent	S					
Enrolment	✓					
Clinical	✓	✓	✓	✓	✓	
Laboratory	✓	✓	✓		✓	
ECG						
Quality of life	S		S	S		
Adverse events						R
Medications						R
Outcome			✓			
Completion				✓		

Arm 2 · drug-resistant TB · 9 months

Form	B	M1	M2	M4	M6	M9	M12	Uns	Log
Screening	✓								
e-Consent	S								
Enrolment	✓								
Clinical	✓	✓	✓	✓	✓	✓	✓	✓	
Laboratory	✓	✓	✓	✓	✓	✓		✓	
ECG	✓	✓	✓	✓	✓	✓			
Quality of life	S				S		S		
Adverse events									R
Medications									R
Outcome						✓			
Completion							✓		

✓ collected · S survey (participant) · R repeating instrument · Uns = Unscheduled (repeating event) · Log = Logs event

TB: tuberculosis · ECG: electrocardiogram

Designing the TBCARE data dictionary

Variable	Form	Type	Label	Choices / calculation	Validation	Branching	Ident.
record_id	screening	text	Study ID				
dob	enrolment	text	Date of birth		date_dmy		y
sex	enrolment	radio	Sex at birth	1, Male | 2, Female			
pregnant	enrolment	yesno	Currently pregnant?			[sex]='2' and [age]>=15	
weight_kg	clinical_assessment	text	Weight (kg)		number_1dp 25–200		
bmi	clinical_assessment	calc	BMI	round([weight_kg]/(([first-event-name][height_cm]/100)^2),1)			
qtcf	ecg_monitoring	text	QTcF (ms)		integer 300–700		
qol_1	quality_of_life	radio	Limited activities?	1, Not at all | … | 5, Completely		matrix: qol	
ae_term	adverse_events	text	Adverse event				

75 fields **11** forms **6** identifiers **5** calculated

File: TBCARE_DataDictionary.csv

ID: identifier · BMI: body mass index · QTcF: corrected QT interval (Fridericia)

Build TBCARE in this order

#	Step	Where	File
1	**Create project**	Practice, empty project	–
2	**Upload data dictionary**	Dictionary > Upload	TBCARE_DataDictionary.csv
3	**Enable longitudinal + surveys**	Project Setup	–
4	**Arms and events**	Define My Events > Upload	TBCARE_arms.csv, TBCARE_events.csv
5	**Designate instruments**	Designate Instruments > Upload	TBCARE_instrument_event_mapping.csv
6	**Repeating**	Unscheduled = event; AE and medications = instruments	set by hand
7	**DAGs**	antwerp, mbarara	set by hand
8	**Surveys**	e-Consent, Quality of life	set by hand

✓ Then **test** with 3 fake records before importing anything.

DAGs: Data Access Groups

Uploading the simulated data

record_id	redcap_event_name	redcap_repeat_instrument	redcap_repeat_instance	redcap_data_access_group	weight_kg	ae_term
1-001	baseline_arm_1			antwerp	61.5	
1-001	month_2_arm_1			antwerp	63.0	
1-001	unscheduled_arm_1		1	antwerp	62.4	
1-001	logs_arm_1	adverse_events	1	antwerp		Nausea
2-003	month_9_arm_2			mbarara	58.2	

Event
redcap_event_name

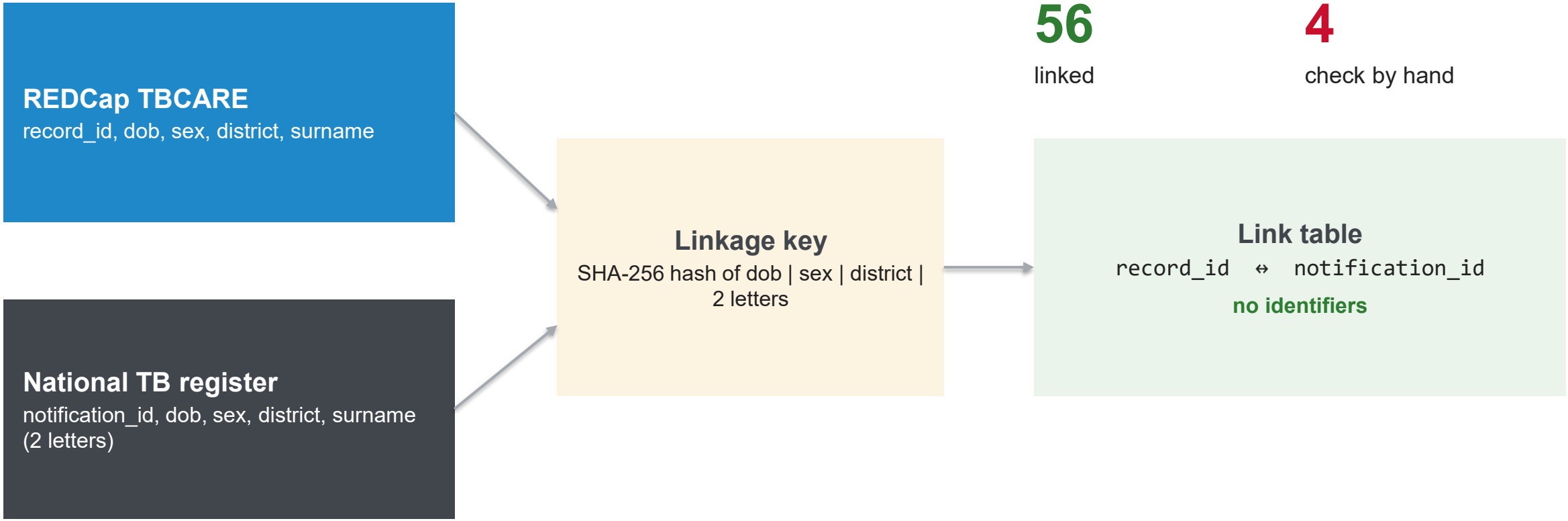
Repeat
instrument + instance

Site
data access group

Not imported: **calculated, file, descriptive** fields. Dates as YYYY-MM-DD; one column per checkbox option.

Files: **TBCARE_simulated_import.csv** (clean, 60 participants) and **…_with_errors.csv** (for the data quality exercise).

Record linking: TBCARE and the national TB register



Linkage uses identifiers: done by the **data manager**, in a secure place, and described in the **DMP**.

TB: tuberculosis · REDCap: Research Electronic Data Capture · SHA-256: a one-way hash function · DMP: Data Management Plan

From REDCap to R with REDCapR

01_redcapr_read.R

```
library(REDCapR)
uri   <- "https://redcap.uantwerpen.be/api/"
token <- Sys.getenv("REDCAP_TBCARE_TOKEN")

ds <- redcap_read_oneshot(uri, token)$data
dim(ds)                    # 439 rows x 88 columns
table(ds$redcap_event_name)

clin <- redcap_read(
  redcap_uri = uri, token = token,
  forms  = "clinical_assessment",
  events = c("baseline_arm_1", "month_2_arm_1"),
  raw_or_label = "label")$data
```

Function	Returns
redcap_metadata_read()	data dictionary
redcap_arm_export()	arms
redcap_event_read()	events
redcap_event_instruments()	form-event mapping
redcap_instrument_repeating()	repeating settings
redcap_write()	import (upload) data
redcap_log_read()	logging (audit trail)



Token in **.Renviron** only. The script can be shared; the token never.

What a longitudinal export looks like in R

record_id	redcap_event_name	redcap_repeat_instrument	#	sex	weight_kg	qtcf	ae_term
2-003	baseline_arm_2			2	54.1	428	
2-003	month_1_arm_2				55.0	441	
2-003	month_2_arm_2				55.6	512	
2-003	unscheduled_arm_2		1		55.2		
2-003	logs_arm_2	adverse_events	1				Nausea
2-003	logs_arm_2	adverse_events	2				Dizziness

```
baseline <- subset(ds, grepl("^baseline", redcap_event_name))
ae       <- subset(ds, redcap_repeat_instrument == "adverse_events")
```

Read it right

One row = record × event × repeat

Empty ≠ missing: form not at that event

sex only at baseline: carry it forward

Each AE is its own row

REDCapDM: queries and missing events

04_redcapdm_queries.R

```
library(REDCapDM)
dataset <- redcap_data(uri = uri, token = token)
tb <- rd_transform(dataset)

q <- rd_query(tb,
  variables  = c("weight_kg", "temp_c"),
  expression = c("x > 200", "x < 34"),
  event      = "baseline_arm_1")
q$results

me <- rd_event(tb, event = "month_6_arm_1")
rd_export(queries = q$queries, path = "TBCARE_queries.xlsx")
```

1. **`rd_transform`**
   labels, checkboxes, branching
2. **`rd_query`**
   find impossible or missing values
3. **`rd_event`**
   who missed a visit?
4. **`rd_export`**
   query report for the sites
5. **`check_queries`**
   what was resolved?



Try it on **TBCARE_simulated_with_errors.csv**: can you find all 7 errors?

REDCapSync and RosyREDCap: standardised pipelines

05_redcapsync_rosyredcap.R

```
library(REDCapSync)
project <- setup_project(
  project_name = "TBCARE",
  redcap_uri   = uri,
  dir_path     = getwd())
project$sync()          # only changed records
project$generate_dataset("custom")

library(RosyREDCap)     # GitHub: thecodingdocs
run_RosyREDCap()
```

REDCapSync

Sync only what changed (uses the log)

Standardised project object

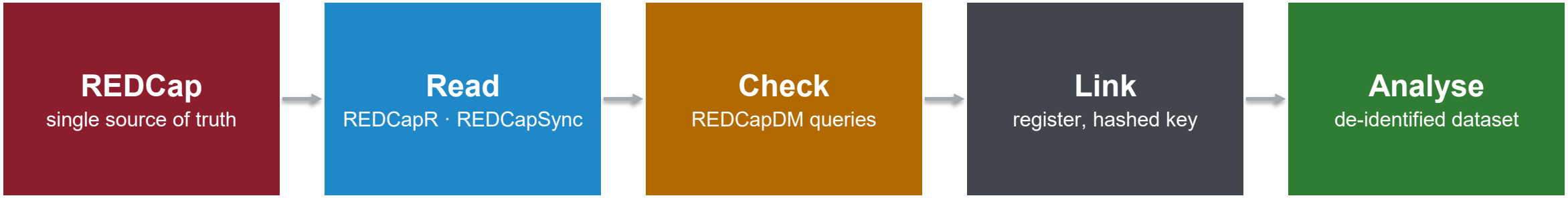
RosyREDCap

Shiny app: explore, de-identify, plot

Not yet for multi-arm projects

Token name: **REDCAPSYNC_TBCARE** in .Renviron.

A reproducible TBCARE pipeline



- Scripts in version control (Git); data and tokens never
- Queries go back to the sites; fixes happen in REDCap, not in R
- Same scripts, same result: write that in the DMP

⚠ Fix data **in REDCap**, never by hand in the exported file.

REDCap: Research Electronic Data Capture · DMP: Data Management Plan

Try it now: R in your browser

1 Load the export
what redcap_read_oneshot returns

2 Check
ranges, impossible values

3 Missing visits
like rd_event

4 Link
national TB register

5 Plot
weight by arm over time

Course-work page

bosangir.github.io/REDCapMepi/coursework

Slides · CRFs · data dictionary · simulated data · R scripts · R in the browser

✓ In the browser: **simulated data only**, nothing leaves your computer.

TB: tuberculosis · CRFs: case report forms

Hands-on lab: build TBCARE in 60 minutes

1. **Create the project** — Practice / Just for fun — slide 9
2. **Upload the data dictionary** — or build Enrolment by hand — slide 14
3. **Add validation** — dates, weight 30–200, required — slide 16
4. **Add branching** — test type if lab done — slide 17
5. **Add BMI** — calculated field — slide 18
6. **Go longitudinal** — 4 events, designate forms — slide 26–27
7. **Add two DAGs** — Antwerp, Mbarara — slide 33
8. **Test and export** — 3 fake records; de-identified CSV — slide 42



Fake data only. Then swap with the next pair and try to break their project.

BMI: body mass index · CSV: comma-separated values

Lab check: did it work?

- **Date of birth:** type 31-02-1990: REDCap should refuse it
- **Weight:** type 615: out-of-range warning
- **Branching:** lab not done: test type stays hidden
- **BMI:** 60 kg, 170 cm gives 20.8
- **Events:** Record Status Dashboard shows 4 columns
- **DAGs:** a Mbarara user cannot see Antwerp records
- **Export:** identifier fields are missing from the de-identified CSV

REDCap: Research Electronic Data Capture · BMI: body mass index · CSV: comma-separated values

Common mistakes, and how to avoid them

Mistake	Better
Real data in Development	Fake data until Production
Free text for sex, district, outcome	Radio buttons or drop-downs
med_1, med_2, med_3 fields	A repeating instrument
Identifier flag not set	Tick Identifier for all personal fields
Renaming variables in Production	Fix names before going live
API token in the script	Token in .Renviron

API: application programming interface

Take-home messages

1. Build in **Development** with fake data; go live only after testing
2. Design the **data dictionary first**: it is also your codebook
3. Fixed lists, validation and branching **prevent** errors
4. Choose the **project type** before collecting data
5. **DAGs**, queries and locking make multi-site work traceable
6. Tick **Identifier** so exports can be de-identified
7. Rewatch any step: projectredcap.org/resources/videos

Resources

REDCap training videos
projectredcap.org/resources/videos (23 videos)

Help and FAQ inside REDCap
always matches your version

Instrument Library
validated instruments, ready to import

UAntwerp REDCap
redcap.uantwerpen.be · ask your administrator

RDM support
rdm-support@uantwerpen.be

Part 2 of this course
the DMP and the TBCARE case study

REDCap: Research Electronic Data Capture · RDM: research data management · DMP: Data Management Plan

Questions?

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